[METRIC]
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October 14, 1994
SUPERSEDING
MIL-S-45351B(AT)
26 March 1984

COMMERCIAL ITEM DESCRIPTION

SEAL, ENGINE SHROUD

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE . This CID covers requirements for engine shroud seal made of treated fiberglass/aramid fabric, stitched over a corrosion resisting wire mesh core, intended for use in vehicles where high temperature conditions are present.

1.1 Classification. Items described in this CID shall be of the following types as specified (see 5.2).

Type I - Round bulb, with the tail extention of the cover extending radially from the periphery of the core material.
Type II - Rectangular bulb, with the tail extention of the cover parallel to the longer side of the rectangular core.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein or in the applicable engineering drawing (AED), the materials used shall be in accordance with the manufacturer's material specifications/drawings. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.4).

2.1.1 Thread. The thread used in stitching the plies of the tail shall be wire reinforced fiberglass with teflon coating and with a minimum breaking strength of 10 kilogram (kg).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-T, Warren, MI 48397-5000.

FSC 2930

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A-A-5249.)

2.1.2 Cover cloth. Cover cloth shall be aramid/glass wire reinforced cloth with silicone rubber on the inside, and coated on the outside with aluminum pigment silicone compound. The base fabric shall be plain weave aramid/glass cloth. Reinforcing wire shall be Inconel or stainless steel of the 300 series and wire diameter shall be American Wire Gage 36 (0.12 millimeter (mm) dia.). The cloth weight shall be 1 kg per square meter (m^2) (maximum). Warp and fillings yarns shall be strands of wire and fiberglass completely covered by aramid fibers. The yarn shall be not less than 50 percent aramid. The wire shall not show in the yarn or cloth. The inner and outer coating shall be silicone rubber Grade 5GE605A19B37E026F19Z of ASTM D2000. The value of Z shall be the minimum adhesion to the base fabric of 2 kg per centimeter (cm) of width.

2.1.3 Cover coatings. When coatings are integrated into the weave of the cloth so that separation of tensile tongue is impossible, adhesion values are assumed to exceed 2.68 kg per cm. The inner and outer coating shall be complete and continuous. No exposed yarns shall be visible through the coating. The coating shall be smooth without blisters or depressions. The cloth after coating shall have the following properties:

a. Thickness: Not less than 1.27 mm.
b. Weight: 1.575 + 0.18 kg per m^2.
c. Tear strength: ‘Not less than 9 kg.

2.2 Design and construction. Seals shall be constructed to conform to the dimensions and form shown in the AED (see 5.2). These seals shall consist of a core of the shape specified (see 1.1) and a cover (see 2.1.2). The cover shall overlap the core. Its edge shall extend to form the tail.

2.2.1 Core of the bulb. The core of the bulb shall be constructed of Inconel alloy or corrosion resisting stainless steel of the 300 series with not less than 16% chromium and 6% nickel. The filler shall be enclosed in a tubular mesh cover of the same material. The configuration and consistency of the filler, diameter ok wire and size of mesh in filler and mesh cover shall be such as to provide the compression and recovery characteristics specified in 2.4, 2.5 and 2.6. When specified in the applicable drawings (see 5.2), the mesh cover shall be drawn over the cut ends of the core, and engaged upon itself in such a way as to form a bound end without distorting the bulb of the seal.

2.2.2 Tail section. The tail section shall be furnished with bolt holes or with cut-outs, as specified in the AED. Sewing shall be of lock stitches and shall count not less than 3 nor more than 5 stitches per running 25 mm. Sewing shall consist of two rows. One stitch line shall be closely adjacent to the core, the other stitch line shall be approximately 6 mm in from the edge of the tail. Bolt holes or cut-outs, when specified, shall not penetrate the stitching adjacent to the bulb.

2.2.3 Length. Length of seals without bolt holes or cut-outs shall not exceed 7.5 meters (m).

2.2.4 Bulb and tail. The dimensions of bulb and tail (overall width) for type I and type II seals shall be as specified in table I.

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TABLE I. Bulb and tail dimensions.

Type	Nominal bulb diameter (mm)	Tolerance on overall width (bulb plus tail) dimension specified in AED
I	9.5 + 1.6	± 1.6
I	12.5 + 1.6	± 1.6
I	16.0 + 1.6	± 1.6
I	25.4 + 3.2	± 2.4
I	38.1 + 3.2	± 3.2
I	44.5 + 3.2	± 3.2
II	Nominal rectangular bulb size 19.1 + 1.6 by 38.1 +3.2	+ 3.2

2.3 Flame penetration. With the bulb of the finished seal compressed to 80 percent of its original nominal size between flat steel plates, the seal shall withstand a 1100 degrees Celsius (°C) flame for 15 minutes with no penetration of flame beyond the sealing point and no flame remaining in the seal for more than 5 seconds after removal of the flame source.

2.4 Load compression. The seals shall be capable of being compressed under load and recovered within the load requirements and recovery limits specified in tables II and III at 24°C and 177°C, respectively. This shall be verified by applying to the type I seals the loads in an axial direction and for the type II seals by applying the loads parallel to the tail and with the tail held down.

2.5 Compression recovery at 24°C. The seals shall recover to the percentage of the original bulb size after 15 minutes subsequent to being compressed for one minute.

2.6 Compression recovery at 177°C. The seals shall recover to the percentage of the original bulb size following each phase when subjected to the test as described below. The seals shall be compressed at the applicable percentage of compression specified in table 111, for 16 + 1/2 hours (phase 1) and for 5 hours (phase 2). The seals shall be allowed to cool for 15 minutes at ambient temperature following each phase. Following 60 + 5 minutes rest, phase 1 and phase 2 shall be repeated for a total of 4 cycle.

2.7 Tear strength. The cover shall exhibit a minimum tear strength of 5 kg before heat test, when tested in accordance with the requirements of ASTM D2261 and D2262.

2.8 Identification markings. Unless otherwise specified in the AED, identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code and part number and the AED part number (see 5.2).

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TABLE II. Load compression.

Type	Nominal bulb size diameter in mm	Load - kg per linear cm of seal			
		Percent compression (%)			
		20%	33%	40%	60%
I	9.5	0.45 to 1.35	1.25 to 2.15	2.15 to 5.00	
I	12.5	0.40 to 1.30	0.94 to 2.20	1.61 to 4.11	
I	16.0	0.36 to 1.25	0.80 to 2.06	1.47 to 3.17	
I	25.4	0.36 to 1.16	0.72 to 2.00	0.90 to 2.70	
I	38.1	0.40 to 0.94		1.03 to 2.10	2.70 to 5.90
I	44.5	0.36 to 0.90		0.98 to 1.88	2.86 to 5.40
	Nominal bulb size rectangular				
II	19.1 by 38.1			2.70 to 8.00	14.30 to 28.60

TABLE III. Percentage recovery.

Type	Nominal bulb size in mm	Percentage (%)		
		Compression	Recovery	
			@24°C	@177°C
I	25.4	33	90	75
I	9.5	33	90	75
I	12.5	33	90	75
I	16.0	33	90	75
I	38.1	40	90	80
I	44.5	40	90	80
II	19.1 by 38.1	40	90	80

3. QUALITY ASSURANCE PROVISIONS.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms

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to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof **of such conformance** prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING. Preservation, packaging, packing, and marking for the desired level shall be as specified in **the** contract (see 5.2).

5. NOTES . (This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. ASTM D2000 '*Standard Classification System for Rubber Products in Automotive Applications", ASTM D2261 "Standard Test Method for Tearing Strength of Woven Fabrics by the Tongue (Single Rip) Method (Constant Rate-of-Extension Tensile Testing Machine)** and ASTM D2262 Standard Test Method for Tearing Strength of Woven Fabrics by the Tongue (Single Rip) Method (Constant Rate-of-Traverse Tensile Testing Machine)" are available from the American **Society** for Testing and Materials (ASTM) , 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Type of seal required.
d. Applicable engineering drawing and revision letter, and part number.
e. Applicable level of packaging, packing, and marking.

5.3 Cross-reference data. Seals, engine shroud conforming to this CID are interchangeable with seals, engine shroud conforming to MIL-S-45351B(AT), dated 26 March 1984.

5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AT

(Project 2930-0015)

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